

Rhino Bank Infrastructure Proposal

North East Intercity Railway

Newcastle – Sunderland – Middlesbrough Rapid Rail Corridor

Creating the First North East Regional Intercity Railway

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Executive Summary

The North East of England contains a population corridor of over **2.5 million people** stretching from:

Newcastle → Gateshead → Sunderland → Hartlepool → Middlesbrough

However, despite being one of Britain's largest urban regions, east-west and coastal rail connectivity remains fragmented.

Current services are constrained by:

- Victorian railway geometry
- slow junctions
- mixed freight/passenger traffic
- bottlenecks around Newcastle and Middlesbrough
- lack of a dedicated fast regional service

Network Rail has identified capacity constraints on the East Coast Main Line around the North East, including the need for interventions such as reopening the Leamside Line to release capacity and improve regional connectivity.

Rhino Bank proposes:

The North East Intercity Railway (NEIR)

A **110 mph regional express railway** connecting:

Newcastle Central

↓

Gateshead

↓

Sunderland

↓

Seaham

↓

Hartlepool

↓

Middlesbrough

A northern equivalent of:

- Crossrail regional services
 - German Regional Express (RE)
 - Swiss InterCity regional corridors
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Strategic Objective

Transform the North East from a series of disconnected towns into a single economic region.

A person living in:

- Sunderland works in Newcastle
- Middlesbrough studies in Newcastle
- Hartlepool accesses Teesside industry
- Newcastle accesses Teesside ports

without requiring a car.

Existing Journey Times

Approximate current rail times:

Route	Current Time
Newcastle–Sunderland	25 mins
Sunderland–Hartlepool	45 mins
Hartlepool–Middlesbrough	35 mins
Newcastle–Middlesbrough	75–90 mins

Problems:

- stopping patterns
 - reversals
 - slow curves
 - congestion
-

Proposed Infrastructure

Phase 1

Sunderland–Middlesbrough Fast Corridor

Cost:

£2.4 billion

Works:

Track upgrades

- 110 mph running
- new passing loops
- higher-speed junctions
- signalling upgrades

Cost:

£700m

Coastal Bypass Sections

Several sections of the current route are slow because the railway follows nineteenth-century coastal alignments.

New infrastructure:

Seaham Bypass

- 8 km new railway
- removes coastal curves

Cost:

£600m

Time saving:

5 minutes

Hartlepool Northern Bypass

- new alignment avoiding slow approaches
- grade separation from freight

Cost:

£700m

Time saving:

8 minutes

Middlesbrough Approach Upgrade

New:

- flyover junction
- tunnelled approaches
- separate passenger/freight routes

Cost:

£400m

Phase 2

Newcastle–Sunderland Fast Link

Cost:

£1.8 billion

Current route:

Newcastle → Sunderland

is shared between:

- Tyne & Wear Metro
- heavy rail
- freight

Proposal:

Dedicated heavy rail express paths.

Works:

- additional tracks
 - junction upgrades
 - signalling separation
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Phase 3

Leamside Regional Railway

Cost:

£1.2 billion

The Leamside Line is strategically important because it can provide additional capacity and alternative routing. Studies have examined reopening the route for passenger and freight use, with previous estimates varying depending on specification.

Rhino Bank proposal:

Reopen as:

- passenger railway
- freight relief route
- future metro connection

Stations:

- Washington
- Penshaw
- Ferryhill
- Durham South

Benefits:

- unlocks Washington
- removes pressure from ECML
- connects Tyne and Tees economies

Complete Network

Total Investment

Component	Cost
Sunderland–Middlesbrough upgrades	£2.4bn
Newcastle–Sunderland improvements	£1.8bn
Leamside reopening	£1.2bn
Digital signalling	£700m
Rolling stock	£900m
Stations	£500m

Total Programme

£7.5 Billion

Train Specification

Fleet:

50 regional express trains

Specifications:

- 110 mph maximum speed
- electric/battery hybrid capability
- regenerative braking
- 4–6 car formations
- WiFi
- workspaces
- metro-style boarding

Capacity:

Approximately:

25,000 passengers/hour peak direction.

New Timetable

Every 15 minutes:

Newcastle ↔ Middlesbrough

Every 10 minutes:

Newcastle ↔ Sunderland

Every 20 minutes:

Middlesbrough ↔ Hartlepool ↔ Sunderland

Journey Time Improvements

Newcastle–Middlesbrough

Current:

75–90 minutes

Future:

45 minutes

Saving:

30–45 minutes

Newcastle–Sunderland

Current:

25 minutes

Future:

15 minutes

Saving:

10 minutes

Sunderland–Middlesbrough

Current:

80 minutes+

Future:

35 minutes

Saving:

45 minutes

Hartlepool–Newcastle

Current:

55–65 minutes

Future:

35 minutes

Saving:

20–30 minutes

Economic Impact

A connected North East effectively creates a single labour market.

Labour Market Expansion

Potential additional accessible workers:

+250,000

GDP Impact

Comparable regional rail investments typically generate benefits through:

- productivity
- commuting access
- land development
- business clustering

Rhino Bank economic model:

Impact	Estimate
Construction employment	35,000 jobs
Permanent railway jobs	4,000
New housing unlocked	50,000 homes
Additional annual GVA	£2.5–3.5bn
30-year economic benefit	£75–100bn

Development Opportunities

Sunderland Waterfront

Potential:

- offices
- housing
- university expansion

Hartlepool

Potential:

- offshore energy
 - port logistics
 - manufacturing
-

Middlesbrough

Potential:

- digital industries
 - chemicals
 - Teesworks expansion
-

Newcastle

Potential:

- finance
 - technology
 - universities
-

Integration With Wider North

Connections:

- Darlington
- York
- Leeds
- Manchester
- Edinburgh

The North East Intercity Railway becomes the eastern branch of a northern economic network.

Rhino Bank Funding Model

Public-Private Infrastructure Partnership

Source	Amount
Rhino Infrastructure Fund	£2.5bn
Pension funds	£2bn
Government	£2bn
Local development capture	£1bn

Final Vision

The North East currently has the population and industrial base of a major European region but lacks the transport infrastructure to function as one.

The Rhino Bank North East Intercity Railway would create:

- a 45-minute Newcastle–Middlesbrough corridor
- a unified Tyne–Tees economy
- faster access to jobs and universities
- increased port competitiveness
- new housing capacity
- a modern regional railway comparable with European metropolitan systems

It would effectively create a "**Northern Crossrail**" for the **Tyne–Tees corridor**, turning Newcastle, Sunderland and Middlesbrough into a single connected economic zone.